

Passages - A Perfect Harmony

6 - It seems that human vocal anatomy isn't as unique as many once believed. We aren't special in the way we make sounds, only in the extent to which we can control those sounds with our brains. Fine neural control over the vocal system is evidently a key factor that separates us from our fellow species.

31 - Whenever two or more different pitches are sounded at the same time, we're in the realm of harmony. Think of music as having two dimensions: horizontal and vertical. Melody consists of single notes played strictly one after another. Together with rhythm it can be thought of as extending horizontally in time. Melody and rhythm can exist perfectly well without harmony and, indeed, much of the world's music throughout history has been essentially non-harmonic.

Harmony forms the vertical dimension - the simultaneous sounding of different pitches - and is what most obviously differentiates Western music from other traditions over the past millennium.

51 - At the heart of the quadrivium was the study of number in various forms: pure number (arithmetic), number in abstract space (geometry), number in time (music) and number in both space and time (astronomy). Following the lead of Pythagoras, Plato saw an intimate connection between music and astronomy: music expressing the beauty of simple numerical proportions to the ears, and astronomy to the eyes. Through different senses, they expressed the same underlying unity based on mathematics.

75 - Over the past hundred years or so, music notation has continued to evolve as composers, now freed from virtually all constraints of tradition, have explored extraordinary new sonic possibilities. So-called graphic notation, in which art and music are combined in a sort of musical map, offers the performer a loose guide, rather than strict instructions, on how to play the music. Graphic scores emerged as a reaction against the rigid constraints of detailed sheet music, using visual symbols, shapes and lines to represent musical ideas. They allow composers to convey concepts like texture, density and spatial relationships between sounds, and provide performers with vastly more interpretive freedom.

85 - The scholastics' aim was to tie reason to faith - to mould the philosophies of ancient Greece to fit a universe in which the overarching truth was the existence of a supernatural God. Above all, scholasticism was a way of teaching, through analytical thinking and intellectual debate, but with the ultimate goal of validating a biblical view of the world. Humanism, as the name suggests, is centred on the nature and importance of humankind. Humanists emphasised personal experience as the starting point for establishing knowledge.

96 - How a musical instrument works is ultimately a matter of physics, and since the laws of physics are the same everywhere, it's not surprising that similar types of instrument have sprung up independently all over the world. A huge variety of drums, flutes and string instruments have been invented, multiple times, in different parts of the globe. All

are based on the idea of causing vibrations in some way that are perceived as being musical or rhythmic. How they've developed is influenced by local factors - the kind of materials to hand and the level of technological skill available to work with them, mythic and other beliefs, and regional patterns of travel and trade. People who live in the frozen north, habitually surrounded by ice and snow, will naturally make instruments from the bone, gut and skin of polar animals. Those who inhabit the tropics will turn to wood, reed and bamboo. Societies that depend on hunting or herding often fashion instruments from the bone and hide of particular animals for spiritual reasons, even when other materials are available.

Absolute or 'perfect' pitch - the ability to reel off any note on the scale at its specific frequency - isn't necessary in situations like these; nor is it important if you're accompanying an instrument, like a guitar, providing you adjust your voice to match the instrument's tuning.

138 - In mediaeval times, churches were the sole venue for the highest forms of music in the West. But large, lofty spaces enclosed by bare stone walls have extraordinarily long reverberation times. Clap your hands inside a fourteenth-century cathedral and it will take several seconds for the sound to die away. Play fast, intricate music in such a place and the effort will be wasted, as the notes bounce off the hard surfaces, this way and that, overlapping and resulting in a cacophony. Even ordinary speech is hard to understand amid all the reverberation. Only one type of music really works in this kind of environment and, not surprisingly, it's the one that came to dominate the

music of the Middle Ages: slow, monophonic chant. Through chant, the monastic choirs of that time could convey sacred texts in a way that took full advantage of the ethereal and spiritual effects produced by the resonance of these cavernous spaces.

140 - Reverberation is the result of lots of reflected waves, which, when they're processed by the brain, are interpreted as a continuous, complicated sound. An echo, on the other hand, is a distinct but quieter copy of the original caused when a pulse of sound comes back to the listener after bouncing off a surface. If there's a delay of more than fifty milliseconds between the first and the second sounds reaching the ear, then they'll be perceived by the brain as separate rather than as one extended event. In terms of distance, an echo can be heard only if the reflecting surface is at least seventeen metres away.

167 - Today, with 12-TET, there's no distinction between the jump in frequency ratio of, say, a major third in one key and a major third in another. But in earlier times, the major third in one key might be close to a perfect fourth in another or a minor third in yet another. A change in key, therefore, could result in a very noticeable shift in the mood of the piece.

185 - Our entire conventional system of tuning in the West is based on the assumption that because an octave interval is consonant in physics it will also sound consonant when perceived by us. But the assumption is flawed.

Musicians and composers like Dolores Catherino, mentioned earlier, are using electronic technology to pioneer polychromatic musical scales with high-

pitch resolution so that notes, intervals and complex harmonies can be tailored to how our hearing actually works. In this way, pitch becomes less of a fixed quantity and more of a dynamic process that puts the listener back at the focus of the music.

186 - Equal temperament, in which all the intervals between adjacent notes are the same, creates a mismatch between the tempered pitches and the embedded harmonics. We've become so accustomed to a relentless diet of 12-TET that we no longer notice that almost every interval played is slightly dissonant and that the harmonics that inevitably arise from tempering are out of balance.

Historically, there was no way around this except by using unequal temperaments. These had the advantage of retaining distinct moods and 'colours' for each key and purer harmonic intonation, but they became increasingly impractical given the way music in the West developed.

202 - Music is the consummate fusion of passion and logic, the yin and yang that all intelligent beings across the universe may experience.

214 - Listening to music you like increases blood flow to the brain more than listening to music you don't.